A same-shower comparison of wavelength shifters and light-yield limits in a compact RADiCAL electromagnetic-shower timing module

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Abstract

Compact calorimeter modules that time electromagnetic showers at shower maximum are candidates for the picosecond-level timing layers of future collider detectors. We compare four builds of the RADiCAL tungsten/LYSO:Ce sampling module — identical except for their wavelength-shifter (WLS) capillaries — in 25–150 GeV electron beams at the CERN SPS: high-light organic DSB1, low-light ceramic LuAG:Ce, a mixed DSB1/LuAG:Ce module, and a multi-segment build. Timing uses a reference-free dual-end differential estimator, with the saturated high-gain rising edge recovered event by event from a linear low-gain copy (srCFD); on identical events this recovery suppresses the non-Gaussian tails that a clipped-peak discriminator develops on saturated pulses. Parametrising $\sigma_t = a/\sqrt{E} \oplus b$, the stochastic term tracks the detected light, growing from $a = 203 \pm 6$ to 440 ± 18 ps $\sqrt{\mathrm{GeV}}$ — more than a factor of two — from the high-light to the low-light build, and the high-light floor of 18.8 ± 0.8 ps confirms the previously published 17.5 ps. The mixed module provides a same-shower control: DSB1 and LuAG:Ce capillaries reading the same showers give per-capillary timing widths agreeing at the 10–20% level ($\sigma_{\mathrm{DSB1}}/\sigma_{\mathrm{LuAG}} = 1.04 \pm 0.05$, scatter-based, under the primary estimator). Set against the cross-build difference, this disfavors intrinsic WLS re-emission kinetics as the dominant explanation, within this geometry and at these light levels: timing performance is governed primarily by detected light yield, module response, and estimator choice. The high-light build reaches 25.7 ± 0.6 ps for the brightest 150 GeV showers (≈ 50 ps over the full fiducial sample), and the mean dual-end asymmetry drifts with energy at -33.6 ± 2.9 ps per e-fold, consistent with shower-maximum migration — supporting evidence that the timing floor carries longitudinal shower information.

Keywords: Fast timing, Sampling calorimeter, LYSO, LuAG:Ce, Wavelength shifter, Silicon photomultiplier, Signal saturation, 4D calorimetry

1. Introduction

Future high-luminosity colliders demand electromagnetic (EM) calorimeters that resolve the time of a shower to a few tens of picoseconds, in addition to its energy, to associate deposits with the correct primary vertex and reject out-of-time pile-up [5, 6, 7, 8, 9]. Dedicated timing layers based on LYSO:Ce with SiPM readout are designed to achieve 30–60 ps per minimum-ionising track [5], and LYSO-based calorimetric devices have demonstrated ~ 30 ps shower timing [13]. The RADiCAL R&D programme develops ultra-compact, radiation-tolerant sampling EM calorimeter modules that time the shower at its maximum, where the signal is largest and most localised [1, 2, 3, 4]: wavelength-shifting (WLS) capillaries

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sample shower maximum and pipe the light to compact SiPM readout at the module ends.

A first RADiCAL test-beam measurement, on a single high-light build, reported a time resolution of ≈ 27 ps at 150 GeV and parametrised the energy dependence as $\sigma_t = a/\sqrt{E} \oplus b$ with $a = 256$ ps $\sqrt{\text{GeV}}$ and a floor $b = 17.5$ ps [1]. That work also identified, as a priority for further study (its Section 7, item 2), a direct comparison of the timing performance of different WLS materials in otherwise-identical modules. This paper presents that comparison.

The question is sharper than "which wavelength shifter is faster." A cross-build timing difference convolves at least five effects that must be disentangled before any statement about the WLS material is meaningful: the detected light yield; the WLS re-emission kinetics (LuAG:Ce is intrinsically slower than the organic DSB1 [16, 17]); the build-to-build module response; the saturation of the timing front end; and the bias of the timing estimator itself on saturated pulses. The distinction matters for detector design: if detected light dominates, the timing of any candidate material is predictable from its light output, and the material can be chosen for radiation hardness or mechanical properties [16] without a separate timing penalty.

This paper contributes, in order: a four-build timing comparison under a single analysis chain (Sections 3–5); an estimator taxonomy in which the saturation-recovered constant-fraction time (srCFD) is primary and fixed-threshold and clipped-peak discriminators are diagnostic, with the residual estimator bias demonstrated on identical events (Section 5.3); a same-shower DSB1/LuAG:Ce control in the mixed module that removes all cross-build systematics by construction (Section 6); a conservative treatment of the timing floors and selection systematics (Section 5.4); and an observed upstream–downstream timing asymmetry whose energy drift is consistent with shower-maximum migration (Section 7).

2. Experimental setup

2.1. Module and builds

The module is a tungsten/LYSO:Ce sampling calorimeter of the "shashlik" type: alternating plates of tungsten absorber and cerium-doped lutetium–yttrium oxyorthosilicate (LYSO:Ce) scintillator, ≈ 25 radiation lengths deep, read out longitudinally by WLS capillaries. Four capillaries traverse the stack and are each read out at both ends, giving $4 \times 2 = 8$ silicon-photomultiplier (SiPM) channels; we label the four downstream ends collectively DW and the four upstream ends UP. The transverse position of each incident electron is measured by a delay-wire chamber [23] and used to define a fiducial region on the beam spot.

Four builds were exposed, differing primarily in the WLS material of the capillaries – and hence in light yield – while sharing the same LYSO:Ce scintillator and tungsten absorber (Table 1): DSB1, high-light organic WLS; LUAG, low-light ceramic LuAG:Ce WLS; MIXED, a single module combining DSB1 and LuAG:Ce capillaries on opposite diagonals; and TENERGY, a multi-segment high-light build with an added longitudinal energy-readout section. DSB1, the brightest, is the build — and the dataset — of Ref. [1]; its timing is re-derived here under the unified analysis chain so that all four builds are compared on an equal footing, and the published results are reproduced for reference (Table 1, Sec. 5.3).

2.2. Readout, beam, and reference

Each SiPM signal is split into a high-gain (HG) copy, optimised for timing, and a low-gain (LG) copy, optimised for energy linearity, digitised by DRS4 switched-capacitor waveform digitisers (CAEN DT5742) [21, 22] over 1024-cell windows with per-cell calibration applied offline: the HG timing channels (and the MCP reference) at 5 GS/s, the LG copies on a second digitiser at 1 GS/s, where they serve as amplitude (not edge-timing) measurements. The HG channels clip at ≈ 820 mV; the LG channels remain linear. Measurements used a 25–150 GeV electron beam at the CERN SPS (May 2023); DSB1 was run at all six energies in 25 GeV steps, the other builds at 50–150 GeV. A microchannel-plate photomultiplier (MCP-PMT) provides a common start; all per-channel times are MCP-referenced, and the differential estimator below cancels this reference.

3. Recovering the saturated edge: the HG/LG-ratio method

Precision timing of all builds on an equal footing requires the same well-defined edge in each, but the HG timing pulses saturate. The recorded HG amplitude piles up at the 820 mV clip (Fig. 1); for all but the smallest showers the waveform is flat across its maximum, and a constant-fraction discriminator (CFD) [19, 20] referenced to the measured peak is forced onto the shallow foot of the pulse, where the slope — and the timing information — is poor.

The LG copy, though too small for precise timing alone, is linear and therefore measures the true amplitude the HG channel would have reached. We use a per-end calibration

$$\mathrm{HG}_{\text{true}} = a_{\text{c}} + b_{\text{c}}\,\mathrm{LG}_{\text{peak}}, \tag{1}$$

fit on the unsaturated population (Fig. 2), to predict the unclipped HG peak event by event. A CFD threshold is then placed at a fixed fraction (15%) of this predicted peak and applied to the recorded HG waveform; because the predicted peak lies far above the clip, the threshold sits on the steep, unsaturated edge. We call the resulting crossing time the saturation-recovered CFD, srCFD (analysis branch name hg_lgcfd); it is the primary timing estimator of this paper, with the fixed-threshold leading-edge time (LED) and the clipped-peak CFD (cfd05) retained as diagnostics.

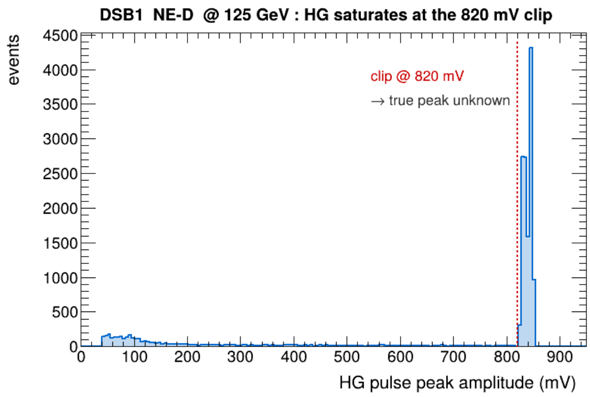

Figure 1: Distribution of recorded high-gain pulse-peak amplitudes (one channel, 125 GeV). The pile-up at ≈820 mV is the digitiser clip: most showers saturate, so the true peak — and the steep edge that leads to it — is not directly recorded.

The recovered edge is a factor ∼ 3.7 steeper than the clipped-peak foot (Fig. 3: 381 vs 102 mV/ns). Since the slew contribution to the time resolution scales as $\sigma_t \simeq \sigma_V/(\mathrm{d}V/\mathrm{d}t)$ with σ_V the voltage noise, a steeper edge directly improves timing. The recovery is attempted uniformly for every build; where the small pulses of the low-light builds (LuAG, TENERGY) defeat the fractional-threshold reconstruction, the per-regime rule of Sec. 5.3 selects a robust fixed-threshold leading-edge time (LED) instead. The choice of per-build timing source is noted in Table 1.

4. Timing estimator and event selection

For each event we form the per-end mean crossing time and combine the two ends into the differential estimator

$$t_{\mathrm{DW-UP}} \equiv \tfrac{1}{2}(\bar{t}_{\mathrm{DW}} - \bar{t}_{\mathrm{UP}})\,, \tag{2}$$

averaging up to four downstream and four upstream SiPM times. The difference cancels any time common to both ends — in particular the MCP start — so $t_{\mathrm{DW-UP}}$ is free of the reference jitter; averaging four channels per end suppresses uncorrelated single-channel noise. The single-event resolution quoted throughout is the standard deviation of a Gaussian fit to the central peak of the $t_{\mathrm{DW-UP}}$ distribution; a small non-Gaussian shoulder is excluded from the fit and has negligible effect on the quoted width. (Ref. [1] noted a satellite of similar ∼ 0.2 ns offset in its MCP-referenced sum distribution and attributed it to a digitiser sampling effect, absent from the differential; the shoulder discussed here appears in the differential under the clipped-peak discriminator and is suppressed by the recovered edge, Sec. 5.3.) The suppression of this structure by srCFD is demonstrated quantitatively in Sec. 5.3 (Fig. B.11).

Events are required to lie in the wire-chamber fiducial region and to have a valid MCP signal. Because shower fluctuations spread the collected light at fixed beam energy, the resolution depends on shower brightness; for a uniform, well-measured comparison across builds we quote the resolution of the 1000 brightest showers (by total low-gain signal) at each energy, which fixes the statistical precision of every point. Within each build, the resolution improves with shower brightness, as expected from the slew argument. For reference, the full fiducial sample at 150 GeV (1.5×10^5 events, no brightness selection) gives ≈50 ps under the same estimator, dominated by its dimmest showers; the dependence on the brightness selection is mapped in Fig. A.9(b) and budgeted in Table 2.

5. Results

5.1. Timing distributions

Figure 4 shows, for DSB1 with all energies pooled and binned by amplitude, the $t_{\mathrm{DW-UP}}$ (top) and absolute (DW+UP)/2 (middle) distributions with their central-peak Gaussian fits, and the fitted σ_t versus amplitude (bottom). The distributions are clean and single-peaked at every amplitude; the differential estimator is consistently narrower than the absolute one, as expected once the common-mode reference jitter is removed, and the width falls monotonically with shower brightness.

5.2. Four-build timing response

Figure 5 is the central energy-scan result. For each build, the brightest-shower σ_t is plotted against beam energy and fit to the published form

$$\sigma_t(E) = \frac{a}{\sqrt{E}} \oplus b, \tag{3}$$

with a the (light-driven) stochastic term and b the floor as $E \to \infty$. The fitted parameters are collected in Table 1. Two features stand out. First, the builds order by light yield, and the stochastic term tracks it: a rises from 203±6 ps $\sqrt{\mathrm{GeV}}$ for high-light DSB1 to 440±18 ps $\sqrt{\mathrm{GeV}}$ for low-light LuAG — more than a factor of two. Second, the floors of the LYSO-based builds lie in the 19–26 ps range: 18.8 ± 0.8 (DSB1), 26.0 ± 1.8 (TENERGY), 24.6 ± 3.3 ps (LuAG). These are mutually consistent once two effects are accounted for: TENERGY times with three capillaries rather than four (a $\sqrt{4/3} \approx 1.15$ statistical penalty, bringing its four-capillary-equivalent floor to ≈22 ps), and the LuAG floor is an extrapolation — its best measured point is 44.4 ps at 150 GeV. The DSB1 floor confirms the 17.5 ps reported in Ref. [1]. (MIXED sits higher, $b = 34.3 \pm 2.4$ ps, for a methodological reason: a single module-wide timing estimator is ill-posed for a module in which half the capillary ends saturate and half do not, so its row in Table 1 is a reference value only; the physically meaningful MIXED comparison is the same-shower analysis of Sec. 6.) The energy dependence is well described by the $1/\sqrt{E}$ (photostatistics) form; a $1/E$ (pure-slew) form fits markedly worse.

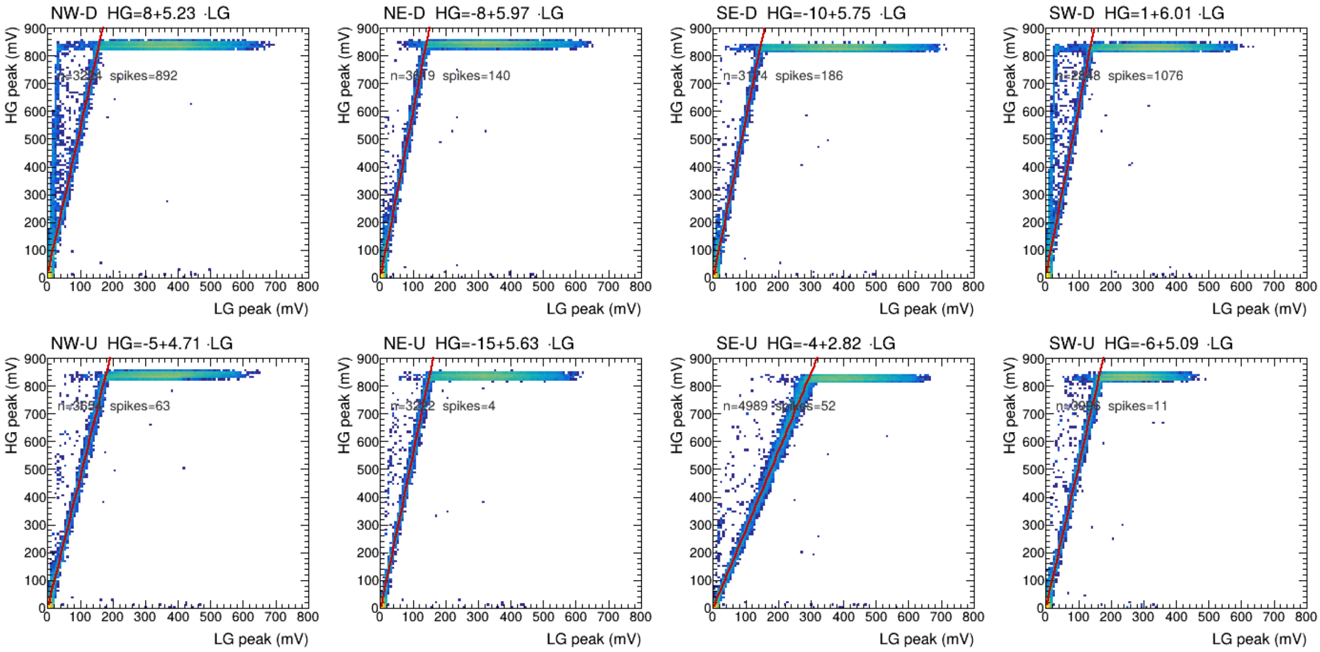

Figure 2: High-gain versus low-gain pulse-peak amplitude for the eight readout channels (100 GeV). The high-gain response saturates at 820 mV (horizontal band) while the low-gain copy remains linear; the per-channel linear fit (red line), Eq. (1), predicts the unclipped true peak.

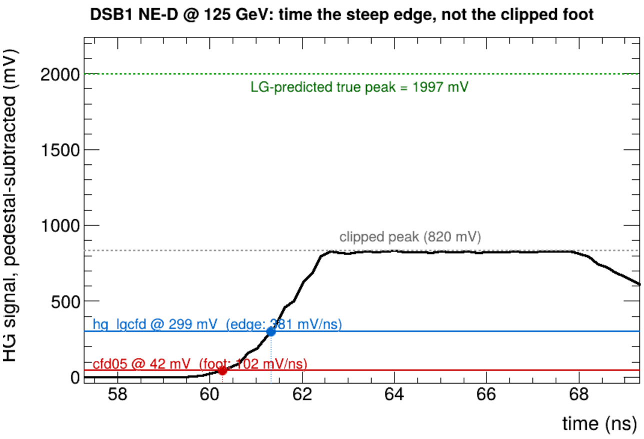

Figure 3: A single saturated high-gain waveform (125 GeV). The recorded pulse clips at 820 mV (grey); the low-gain copy predicts a true peak of 1997 mV (green). The CFD at 15% of the predicted peak (hg_lgcfd, blue) times an edge of 381 mV/ns, against 102 mV/ns at the clipped-peak foot where the 5% CFD (cfd05, red) must sit — a factor 3.7 steeper.

The interpretation: detected light yield governs the dominant, material-dependent (stochastic) part of the time resolution, while the fitted floors are consistent within their uncertainties and caveats. A lower-light configuration times worse in proportion to its reduced light — the behaviour anticipated by the programme's GEANT4 studies, which set the expected single-module time resolution by the detected light level [1, 3]. Whether the residual cross-build difference reflects the wavelength-shifter species (its re-emission kinetics) or only its light output

Table 1: Per-build fit of $\sigma_t = a/\sqrt{E} \oplus b$ to the brightest-shower resolution, the timing source used, and the resolution measured at 150 GeV (statistical errors; selection systematics in Table 2). The WLS column gives the wavelength-shifter material of the capillaries; the LYSO:Ce scintillator and tungsten absorber are common to every build. The final row reproduces the published DSB1 result of Ref. [1], obtained there with a threshold discriminator on the clipped high-gain leading edge; cfd05 is the present chain's closest member of that estimator class and reproduces the published values on this dataset. TENERGY carries a $\sqrt{4/3}$ capillary-count penalty; the MIXED module-wide row is a reference value (single-estimator timing is ill-posed there, Sec. 6).

Build	WLS	src	a (ps $\sqrt{\text{GeV}}$)	b (ps)	σ_t^{150} (ps)
DSB1	organic, hi	srCFD	203 ± 6	18.8 ± 0.8	25.7 ± 0.6
TENERGY	organic, seg	LED	198 ± 21	26.0 ± 1.8	30.3
MIXED	org.+cer.	srCFD	253 ± 29	34.3 ± 2.4	39.6
LUAG	ceramic, lo	LED	440 ± 18	24.6 ± 3.3	44.4
DSB1 [1]	organic	cfd05	256	17.5	27

cannot, however, be decided from the energy scan alone — a between-build comparison cannot separate the two. The mixed module provides the controlled, same-shower test (Sec. 6).

5.3. The gain from recovering the edge

The benefit of the HG/LG-ratio method is isolated by a direct comparison on identical showers (Fig. 6): the brightest-1000 DSB1 resolution is computed with cfd05 (a CFD on the clipped measured peak, the clipped-peak discriminator class of Ref. [1]) and with srCFD (the recovered edge; hg_lgcfd), on an identical-event design: events enter only if valid under every estimator, and the brightest 1000

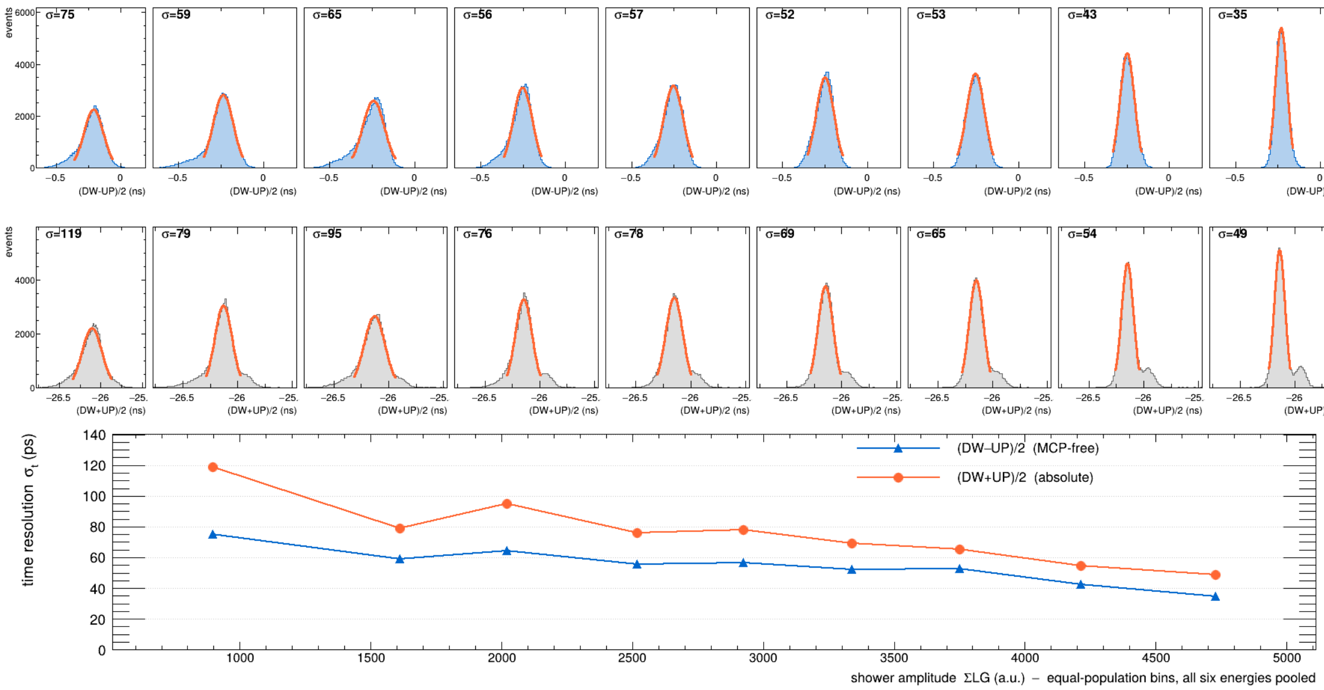

Figure 4: DSB1 timing distributions, all energies pooled and binned by amplitude. Top: (DW − UP)/2 (MCP-free). Middle: (DW + UP)/2 (absolute). Each σ (ps) is the central-peak Gaussian fit. Bottom: the fitted σ_t versus amplitude. The peaks are clean Gaussians at every brightness.

are selected once, so all estimators time the same showers. Two width conventions answer two questions. On the Gaussian core (the convention of every resolution in this paper), srCFD improves $\sigma_t(150)$ by 1.3 ps (26.9 → 25.6 ps, 5%); on a tail-sensitive truncated-RMS width the improvement is 4.0 ps with a paired-bootstrap 68% interval of $[3.4, 4.7]$ ps — the difference is the non-Gaussian shoulder (the ∼ 0.2 ns satellite of Sec. 4) that cfd05 develops on clipped pulses and srCFD removes. The gain is energy-dependent and follows the mechanism: +5.2 ps at 25 GeV (the 5% foot of a small pulse sits in noise), ≈ 0 at 50–75 GeV (moderate clipping, healthy foot), growing again to +2.7 to +3.6 ps at 100–125 GeV, and concentrated in saturated events at 150 GeV (fully clipped subset: 27.4 → 25.3 ps). The fitted floor moves from 21.8 ± 2.7 ps (cfd05, a poor, non-monotonic fit) to 19.9 ± 0.8 ps (srCFD, well behaved) — consistent within errors; the method's value is as much the quality of the fit as its central value.

The tail character of the gain is shown directly in Fig. B.11. In a common window $|x - \mathrm{med}| > 66$ ps (2.5σ of the srCFD core), the tail fraction of the same 1000 events drops from 4.60% (cfd05) to 3.30% (srCFD; 46 → 33 events; Gaussian expectation 1.24%), while the robust IQR widths are nearly identical (28.7 vs 28.3 ps): the cfd05 excess lives in the non-Gaussian tails, and the srCFD remainder above the Gaussian expectation reflects genuine shower tails that no estimator removes. At 75 GeV, where clipped and less-clipped populations coexist, the cfd05 tail excess concentrates in clipped events (3.39% vs 1.95% for srCFD) and largely vanishes in unclipped ones — where, conversely, srCFD's low-gain amplitude anchor is noisy on dim pulses and its own tail fraction rises (7.15%). Each estimator is biased in the regime it was not designed for; this is the basis of the per-regime adopted-source rule (srCFD for bright/clipped, fixed-threshold for dim). Method choice is therefore a correction of estimator bias, not a new detector effect, and does not alter the four-build conclusions of Sec. 5.2.

(The stochastic terms in this controlled comparison reflect the identical-event selection and differ from Table 1, which uses each build's own validity; holding the events fixed isolates the timing method.) The best single-event resolution measured, for the brightest 150 GeV DSB1 showers, is 25.7 ± 0.6 ps; over the full fiducial sample, without the brightness selection, the same estimator gives ≈50 ps at 150 GeV (Sec. 4).

5.4. Systematic uncertainties

The headline brightest-1000 (DW − UP)/2 resolution was recomputed under deliberate variations of every selection knob, on the adopted timing source: the brightness selection ($K = 500$ and 2000 versus the nominal 1000), the timing fiducial radius (2.5 and 3.5 versus 3.0 mm), the MCP saturation window (upper cut 700 versus 750 mV), the per-channel HG threshold (30 versus 20 mV), and the in-event consistency-veto window (1.5 and 3.0 versus 2.0 ns); the fit-range dependence of the floor is reported in the floor block of Table 2. The shifts are collected in Table 2, with

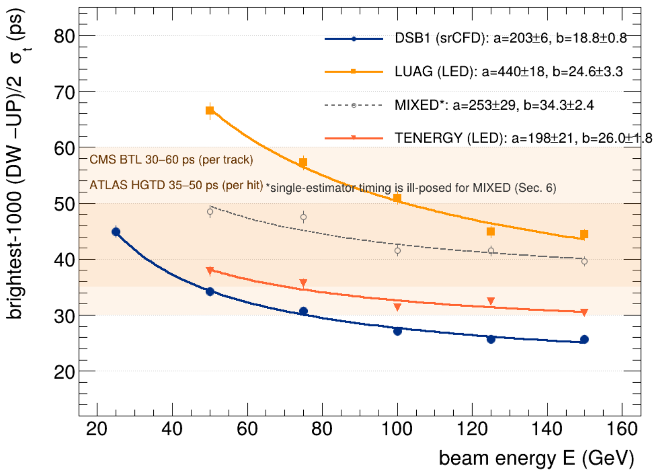

Figure 5: Brightest-shower (DW − UP)/2 time resolution versus beam energy for the four builds, each fit to $\sigma_t = a/\sqrt{E} \oplus b$ [Eq. (3)], using the production estimator chain. The stochastic term a tracks light yield (DSB1 lowest, LuAG highest). The MIXED module-wide curve is a reference only (see text and Sec. 6).

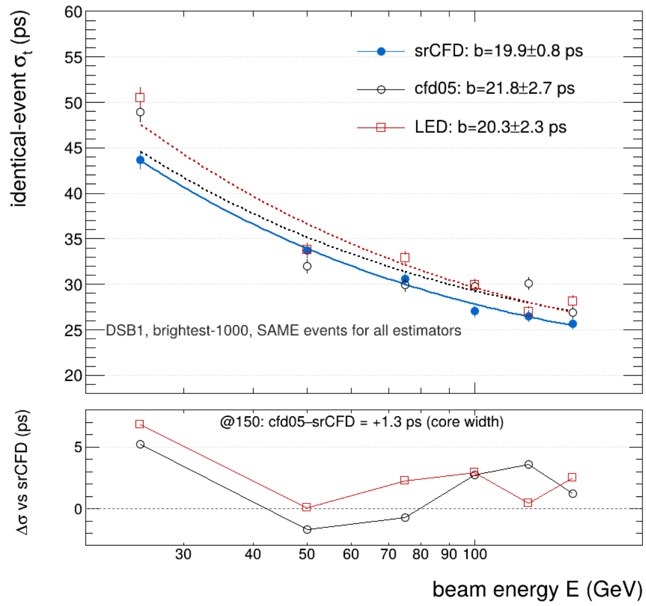

Figure 6: Identical-event comparison of the timing estimators for DSB1 (brightest-1000, production selection): srCFD (recovered edge, filled), cfd05 (clipped-peak CFD, open circles), and LED (open squares), with $a/\sqrt{E} \oplus b$ fits (top) and the difference to srCFD (bottom). At 150 GeV the core-width gain over cfd05 is 1.3 ps; the tail-sensitive gain is 4.0 ps (68% CI $[3.4, 4.7]$, paired bootstrap).

the total systematic taken as their RMS (fit-parameter errors carry the standard $\sqrt{\chi^2/\mathrm{ndf}}$ scale factor [25]; paired uncertainties use the event bootstrap [26]). The selection values themselves were fixed by the scans of Appendix A (Fig. A.9); the timing-method choice (cfd05 versus the recovered edge) is the subject of Sec. 5.3 and is not folded into this budget.

The headline is robust: the total selection systematic on $\sigma_t(150)$, recomputed with the production chain on identical stored events, is ± 1.0 ps (DSB1), ± 1.1 ps (TENERGY), ± 0.9 ps (MIXED), and ± 1.9 ps (LUAG), the last driven by its fiducial sensitivity. The post-fix budget adds the in-event veto window as a knob (1.5 and 3.0 versus the nominal 2.0 ns); its largest effect is -2.3 ps on TENERGY. The largest generic variation remains loosening the brightness selection to $K = 2000$, which admits dimmer showers and genuinely degrades the resolution by 1.4–1.7 ps — a manifestation of the same light–resolution dependence that is the paper's subject, not an analysis artefact. Crucially, the DSB1 floor is stable: $b = 18.8 \pm 0.8$ ps with a fit-range (drop 25 GeV) shift of only $+0.2$ ps, so its agreement with the published 17.5 ps survives the budget. The elevated floor of the MIXED module-wide fit is the ill-posed-estimator feature discussed in Secs. 5.2 and 6.

6. The same-shower control: DSB1 versus LuAG:Ce in the mixed module

The energy scan of Sec. 5.2 cannot by itself distinguish whether the low-light build times worse because of its wavelength-shifter species (slower re-emission kinetics) or because of its reduced light. The MIXED module breaks this degeneracy: two of its four capillaries carry DSB1 and two carry LuAG:Ce, on opposite diagonals, so both materials read the same showers in the same LYSO:Ce scintillator, the same module, and the same digitiser group. The corner–material assignment is confirmed by amplitude-independent pulse-shape discriminants (low-gain charge-to-peak ratio and constant-fraction rise time) validated on the single-material modules; all eight capillary ends classify consistently.

For each event with all timing ends valid, we form the in-event difference of the two same-material downstream ends — $t_{\mathrm{NE}} - t_{\mathrm{SW}}$ for DSB1 and $t_{\mathrm{NW}} - t_{\mathrm{SE}}$ for LuAG:Ce — whose widths measure the per-capillary timing jitter of each material with beam arrival time, group timebase, and the MCP reference cancelling by construction. Figure 7 shows the result under the primary srCFD estimator. The per-capillary widths of the two materials agree at the 10–20% level at every energy; the width ratio $\sigma_{\mathrm{DSB1}}/\sigma_{\mathrm{LuAG}}$ is 0.94, 1.06, 0.93, 1.07, and 1.19 at 50–150 GeV, with a five-energy mean of 1.04 ± 0.05, where the uncertainty is the energy-to-energy scatter (which dominates the per-point statistical uncertainty). The ratio is estimator-dependent at a comparable level — 1.04 (srCFD), 1.16 (cfd05, whose clip-walk penalises the saturating DSB1 corners), 0.80 (LED, whose fixed threshold penalises the dimmer LuAG corners) — so srCFD, the least clipping- and walk-sensitive choice, defines the headline, and the others are diagnostic. Two caveats apply: the pair widths contain a common transverse position-coupling term that dilutes the ratio toward unity (its magnitude is not yet quantified; the strong estimator dependence above shows the widths nonetheless

Table 2: Selection-systematic budget for the brightest-1000 (DW − UP)/2 time resolution at 150 GeV, recomputed with the production estimator chain. Each row is the shift of $\sigma_t(150)$ under one selection variation on identical events; the total is the RMS of the shifts. The floor block gives the DSB1 fit floor and its fit-range stability. The timing-method choice (srCFD vs. cfd05/LED) is treated in Sec. 5.3 with its own paired uncertainty and is not folded in here. Notes: the MIXED module-wide estimator is ill-posed (Sec. 6; its same-shower ratio carries ±0.05 energy-period scatter and < 0.01 run-jackknife spread); TENERGY times with three capillaries ($\sqrt{4/3}$ statistical penalty); LuAG and TENERGY floors are extrapolations (Sec. 5.2).

Variation	DSB1	TENERGY	MIXED	LUAG
shift in $\sigma_t(150)$ [ps]				
K=500	−1.7	−1.4	−0.2	+2.4
K=2000	+1.7	+1.7	+1.4	−1.0
r_{fid}=2.5 mm	+1.2	−0.6	−0.0	−2.8
r_{fid}=3.5 mm	−0.5	+0.4	+0.7	+3.9
MCP <700 mV	−0.3	−0.2	+1.1	−0.3
HG ≥30 mV	+0.0	+0.0	+0.0	+0.0
veto 1.5 ns	−0.5	−2.3	+0.1	+0.0
veto 3.0 ns	−0.2	+0.0	+1.5	+0.0
nominal $\sigma_t(150)$	25.7	30.3	39.6	44.4
total syst.	±1.0	±1.1	±0.9	±1.9
DSB1 floor b (stat)	18.8 ± 0.8 ps			
DSB1 b, fit 50–150 only	+0.2 ps shift			

retain per-capillary sensitivity), and the comparison is specific to this geometry and these light levels.

This is the configuration in which kinetics would be expected to show if they mattered: LuAG:Ce is intrinsically slower than DSB1 [16, 17], and in photostatistics-limited timing the resolution scales with both the photon count and the emission time profile [14, 15]. Set against the more-than-factor-of-two stochastic-term difference between the pure builds, the same-shower near-parity disfavors intrinsic wavelength-shifter re-emission kinetics as the dominant cause of the cross-build timing difference. It does not establish universal material equivalence: at lower light levels or in other geometries the kinetics could re-emerge.

7. Longitudinal timing asymmetry

The (DW − UP)/2 estimator is, by construction, sensitive to the longitudinal position of the light emission: a deeper shower is closer to the downstream ends, shifting the asymmetry negative. Shower maximum migrates downstream as $\sim X_0 \ln E$ [24, 25], predicting a drift of the mean asymmetry of $-X_0/v_{\text{eff}} \approx -26$ ps per e-fold of energy for axial light propagation. The measured drift, on the full fiducial sample with the amplitude-independent srCFD estimator, is monotonic and negative with slope −33.6 ± 2.9 ps per e-fold — consistent in sign and magnitude with the shower-maximum migration (the ∼ 30% excess over the axial estimate is consistent with modal dispersion lowering the effective propagation speed). The drift is read only by constant-fraction estimators; fixed-threshold times, which fire on the earliest photons, suppress it. The slope is an acceptance-weighted population quantity (the fixed-position wavelength-shifter filament samples different shower depths with different efficiency), so we use it here only as a consistency reading: the event-to-event spread of the same asymmetry is the timing floor of Sec. 5.2, and its mean drift confirms that the floor carries genuine longitudinal shower information rather than instrumental jitter alone. The calibration of this asymmetry as a longitudinal coordinate is deferred to a companion paper on the energy and position response of the module.

8. Discussion

These measurements establish detected light yield as the primary design lever for the timing of RADiCAL modules: the cross-build stochastic term tracks the light, and the same-shower control of Sec. 6 disfavors wavelength-shifter kinetics as the dominant alternative explanation, within this geometry and at these light levels. On this evidence a WLS material may be selected for radiation hardness, mechanical robustness, or cost, with its timing then estimated from its light output. LuAG:Ce, attractive for its radiation hardness [16], times worse than DSB1 at a level commensurate with its lower light yield; the same-shower comparison bounds any additional species-specific penalty at the 10–20% level under the primary estimator. As an engineering extrapolation (not a measurement): recovering part of the measured factor-of-∼ 3 light deficit (LuAG:Ce capillary amplitudes are roughly one-third of DSB1's in the mixed module) — for example by doubling the LYSO:Ce thickness in the shower-maximum region, as already foreseen in Ref. [1] (its Section 7, item 3), which doubles the detected light and, with $a \propto 1/\sqrt{N_{\text{phot}}}$, gives $a \to a/\sqrt{2} \approx 310$ ps $\sqrt{\text{GeV}}$ — would bring $\sigma_t(150)$ to ≈ 31–35 ps (depending on the realised floor), i.e. the radiation-tolerant configuration would enter the regime of the dedicated collider timing layers (30–60 ps per track for the CMS barrel timing layer [5]; ∼ 35–50 ps per hit for the ATLAS HGTD [6]) while measuring showers, not single tracks — on eight SiPMs (sixteen readout channels) in a 14×14 mm^2 cell. Among per-shower devices, the W+garnet-fibre SPACAL prototype reaches (18.5 ± 0.2) ps at 5 GeV [10] and the Crilin crystal cells reach < 25 ps above 3 GeV of deposited energy [11, 12]; Table 3 situates these systems, which differ in timed object, energy range, and scale, without ranking them.

The fitted floors of the LYSO builds lie in the 19–26 ps range, consistent within their uncertainties and the stated caveats, and the DSB1 floor confirms the previously published value. The floor's origin is plausibly the shower itself: the (DW − UP)/2 estimator is sensitive to the longitudinal position of energy deposition, and the measured energy drift of its mean (Sec. 7) is consistent with the shower-maximum migration — so shower-to-shower depth fluctuation produces an irreducible spread that does not decrease with light and is common to any scintillator filling

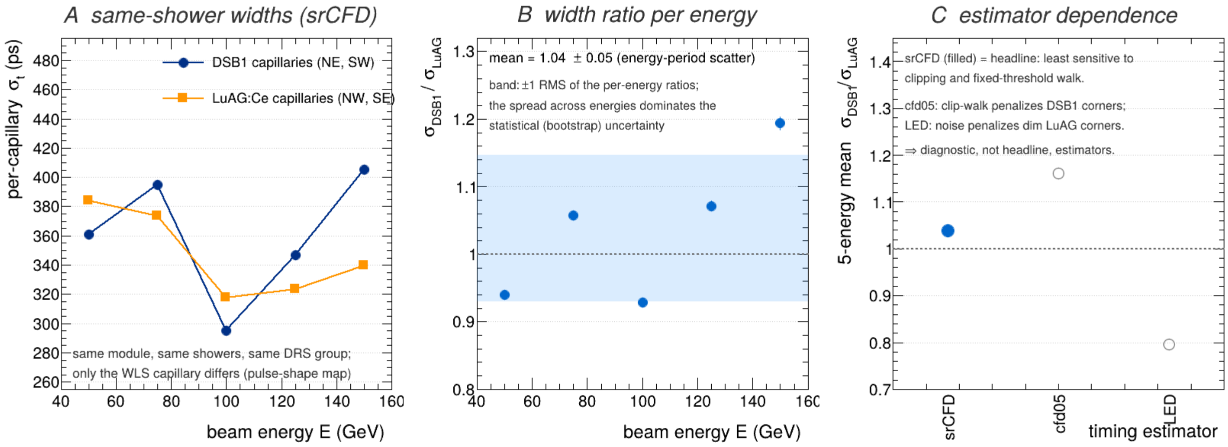

Figure 7: Same-shower comparison in the MIXED module (corner map confirmed by pulse-shape discriminants). (A) Per-capillary timing widths versus energy for the DSB1 (NE, SW) and LuAG:Ce (NW, SE) capillary pairs under srCFD; widths are same-layer in-event pair differences divided by $\sqrt{2}$, so common-mode terms cancel. (B) The width ratio per energy with the ±1 RMS energy-scatter band; the scatter dominates the statistical uncertainty. (C) Estimator dependence (srCFD headline; cfd05 and LED diagnostic).

Table 3: Representative fast-timing systems for context. Design values are from technical design reports; prototype values are published beam-test results. Per-track (MIP), per-hit, per-cell, and per-EM-shower timing are distinct objects and are not directly interchangeable; energies and selections also differ. This table situates the present result; it does not rank.

System	Timing element	Object timed	Characteristic result	Scale	Comparability
This work (RADiCAL, DSB1)	W/LYSO:Ce + WLS capillaries, SiPM	EM shower at shower max	25.7 ± 0.6 ps (150 GeV, brightest sel.); floor 18.8 ± 0.8 ps	8 SiPMs (16 channels) / $14{\times}14$ mm^2 cell	per shower; self-referenced differential
CMS BTL [5]	LYSO:Ce bars + SiPM	MIP track	30–60 ps per track (begin–end of life, design)	barrel layer, $\mathcal{O}(10^5)$ channels	per MIP; design values
ATLAS HGTD [6]	LGAD silicon	MIP hit	≈35–50 ps per hit (design)	$\mathcal{O}(10^6)$ channels	per hit; silicon, design values
LHCb SPACAL R&D [10]	W + garnet crystal fibres, PMT	EM shower	(18.5 ± 0.2) ps at 5 GeV (direct coupling; ∼30 ps with light guides)	prototype cell	per shower; 1–5 GeV beam
Crilin [11, 12]	PbF_2 crystals + SiPM	per-cell deposit	< 100 ps for > 1 GeV (Proto-0); < 25 ps for > 3 GeV (follow-up)	two-crystal prototype	per cell deposit

the same geometry. This same depth sensitivity suggests the route below the floor: a depth-corrected estimator, using the dual-end information to estimate and subtract each shower's depth, which we are pursuing.

9. Conclusions

Three results summarise this work. First, the best timing performance of the RADiCAL module is $\sigma_t = 25.7 \pm 0.6$ ps for the brightest 150 GeV showers of the high-light DSB1 build (≈50 ps over the full fiducial sample without the brightness selection), with $\sigma_t(E) = a/\sqrt{E} \oplus b$, $a = 203 \pm 6$ ps $\sqrt{\text{GeV}}$ and $b = 18.8 \pm 0.8$ ps — the floor confirming the previously published 17.5 ps. Second, while the cross-build stochastic term grows by more than a factor of two from the high-light to the low-light build ($a = 440 \pm 18$ ps $\sqrt{\text{GeV}}$ for LuAG:Ce), the same-shower comparison inside the mixed module gives per-capillary timing widths agreeing at the 10–20% level ($\sigma_{\text{DSB1}}/\sigma_{\text{LuAG}} = 1.04 \pm 0.05$ under the primary estimator): wavelength-shifter re-emission kinetics alone are therefore unlikely to dominate the cross-build difference, within this geometry and at these light levels. Third, the practical design lessons are to maximise detected light, to control saturation and clipping (or recover from them, as the srCFD estimator does by suppressing clipping-induced tails), and to choose timing estimators by regime rather than universally. Future work will calibrate the upstream–downstream timing asymmetry — whose energy drift is consistent with shower-maximum migration — as a longitudinal coordinate, and will optimise the module geometry to recover the light deficit of the radiation-tolerant configuration.

Appendix A. Selection optimization and systematic stability

The four choices that define the headline estimator were each fixed by a scan at the headline energy (150 GeV), shown for all builds in Fig. A.9. Panel (a) compares timing sources: a constant-fraction time on the clipped peak, swept over fractions 3–50% (solid), against the recovered-edge source we adopt (srCFD/LED, dashed). The adopted source lies below the entire clipped-peak family for every build — dramatically so for the low-light LUAG —

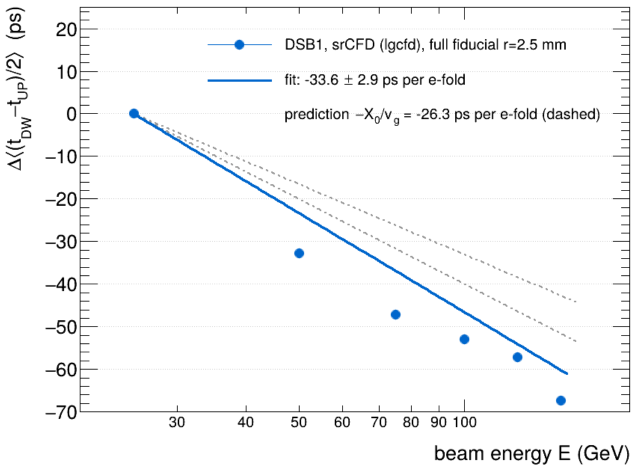

Figure 8: Energy dependence of the mean dual-end timing asymmetry $\Delta\langle(t_{\rm DW} - t_{\rm UP})/2\rangle$ (DSB1, srCFD, full fiducial), relative to 25 GeV. The dashed band is the $-X_0/v_g$ expectation from shower-maximum migration. The fitted slope is -33.6 ± 2.9 ps per e-fold (acceptance-conditional; see text).

which is the quantitative basis for the method of Sec. 3. Panel (b) shows the brightness selection: σ_t falls steeply as the dimmest showers are removed and flattens near $K \sim 1000$, beyond which admitting dimmer showers degrades it again; $K = 1000$ sits at the knee. Panel (c) shows the resolution is flat across the timing fiducial radius out to the chosen 3.0 mm, rising only as off-axis showers leak in. Panel (d) shows insensitivity to the per-channel HG threshold, confirming the 20 mV working point is well clear of the noise floor. The corresponding systematic stability of $\sigma_t(150)$ under these variations is summarised in Fig. A.10 and Table 2.

Appendix B. Tail composition of the method gain

Figure B.11 shows the identical-event tail diagnostics supporting Sec. 5.3: the median-centred $(\mathrm{DW} - \mathrm{UP})/2$ distributions at 150 GeV, the width and tail metrics, and the 75 GeV clipped/less-clipped split including the dim-pulse reversal that motivates the per-regime adopted-source rule.

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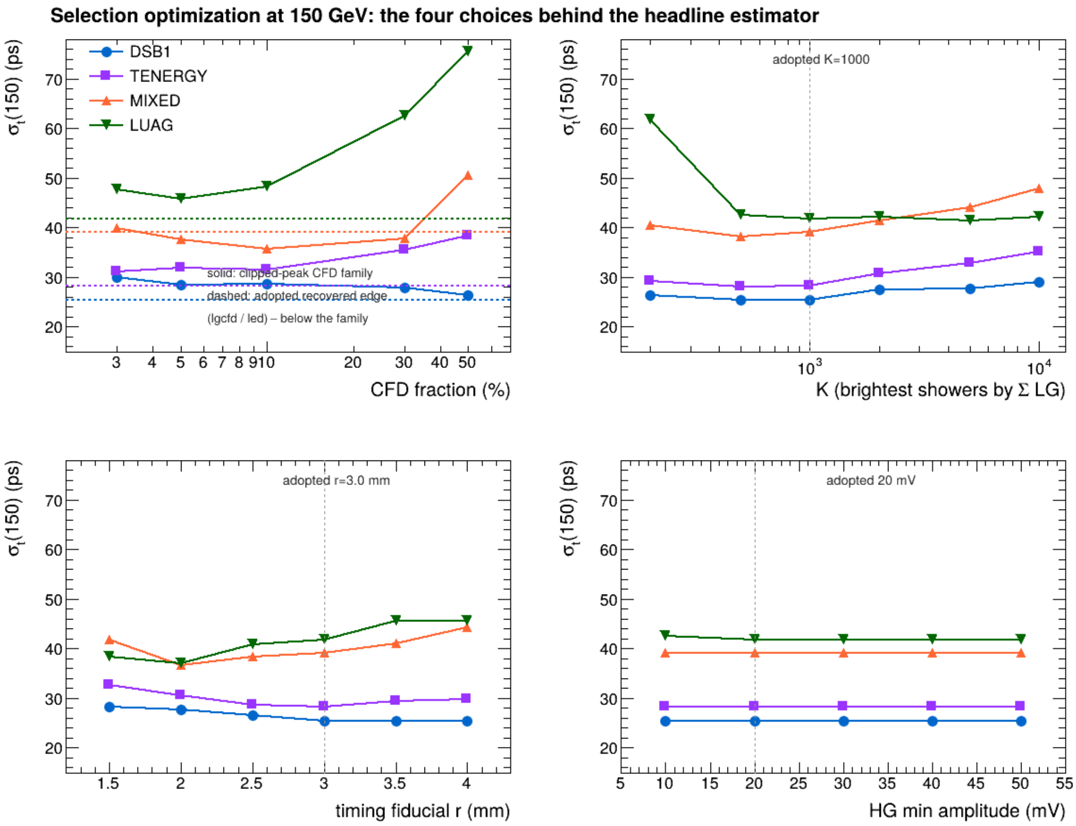

Figure A.9: Selection optimization at 150 GeV for all four builds: (a) timing source as a function of CFD fraction on the clipped peak (solid) with the adopted recovered-edge source overlaid (dashed); (b) resolution versus the brightest-N selection K; (c) versus timing fiducial radius; (d) versus the per-channel HG threshold. Dashed grey lines mark the adopted working points.

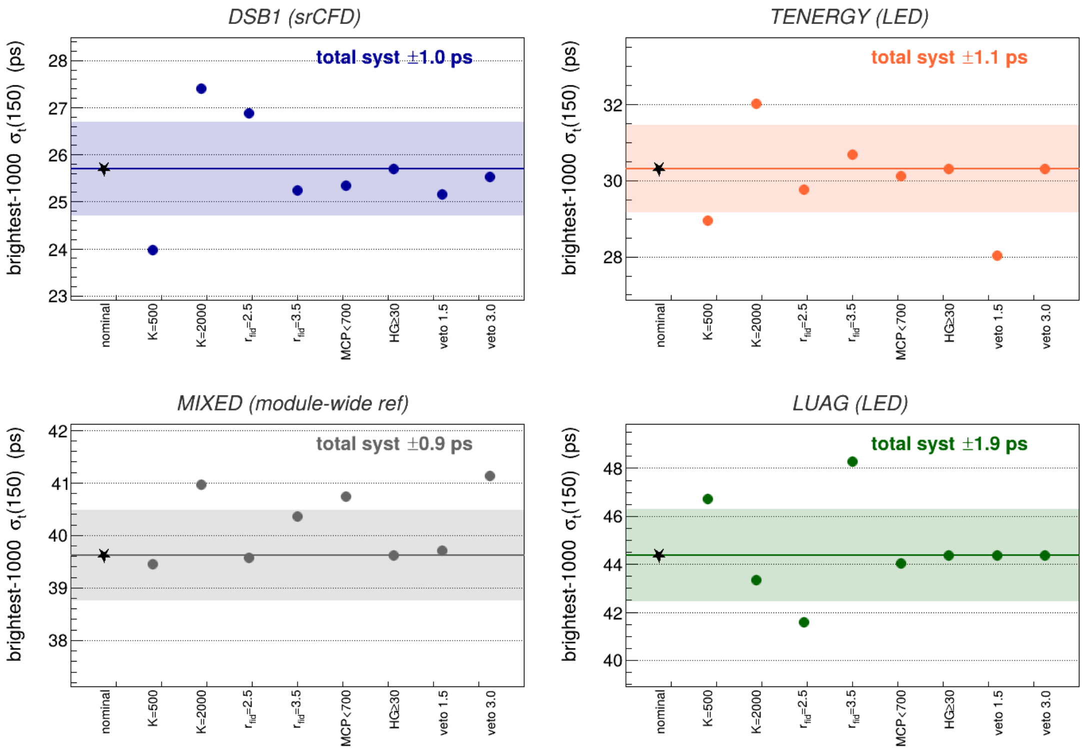

Figure A.10: Systematic stability of the brightest-1000 $\sigma_t(150)$ for each build under the production estimator chain: the nominal value (star) with the total-systematic (RMS) band of Table 2, and each selection variation of Table 2 as a point, including the in-event veto-window variations. The annotated total is the RMS of the shifts; authoritative values are those of Table 2 (floors: Table 1). The largest excursions are the brightness-selection variations (K).

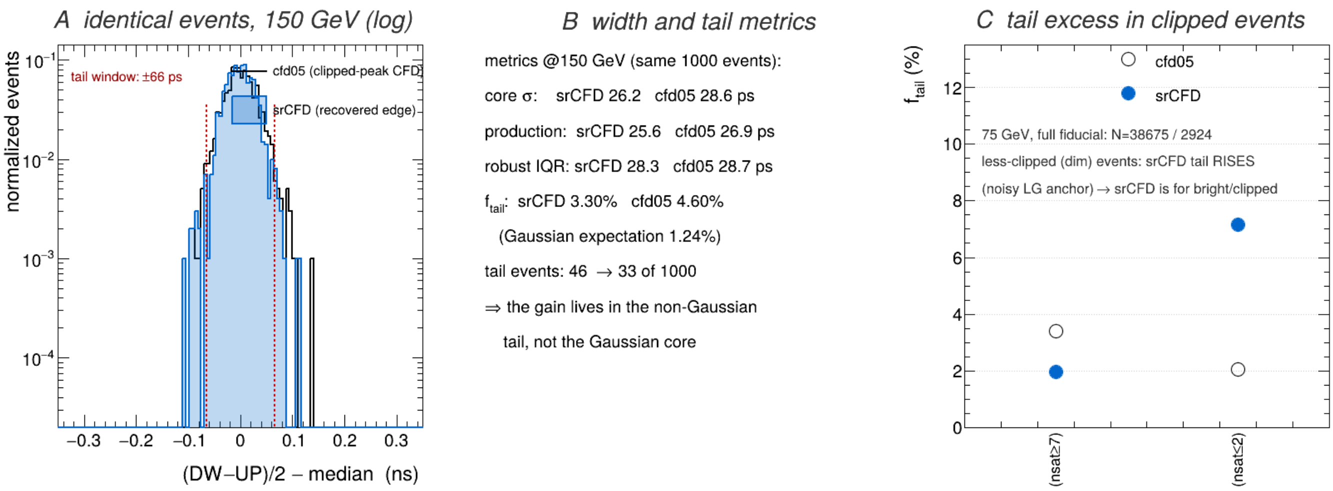

Figure B.11: Tail suppression by the saturation-recovered CFD on identical events (DSB1, brightest-1000, production selection). (A) The per-event (DW − UP)/2 distribution at 150 GeV under cfd05 (line) and srCFD (filled), median-centred, unit-normalized, logarithmic scale; the common tail window $|x - \mathrm{med}| > 66$ ps (2.5σ of the srCFD core) is marked. (B) Width and tail metrics on the same events: the production widths are 25.6 (srCFD) vs 26.9 ps (cfd05) while the tail fraction drops from 4.60% to 3.30% (Gaussian expectation 1.24%; the srCFD remainder reflects genuine shower tails). (C) At 75 GeV (full-fiducial sample; in the brightest-1000 the less-clipped subset is empty), the cfd05 tail excess concentrates in the clipped subset, while on dim unclipped pulses srCFD's low-gain anchor is noisy and its tail rises — each estimator is biased outside its design regime, the basis of the per-regime source rule.